Rudranarayan Muduli

Software Engineer

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Strong in design and integration with intuitive problem-solving skills. Ability to translate requirements into technical solutions with customer satisfaction. Eager to tackle more complex problems and continue to find ways to maximize user efficiency.

EXPERIENCE

(05/2023 - Present) **Backend Engineer - Kemsys Technologies PVT. LTD.**

HRMS Platform (Backend Project)

- Built a full-featured, enterprise-grade HRMS platform similar to GreytHR and Zoho People, covering employee lifecycle, roles & permissions, attendance/timesheets, leave management, and project allocations.
- Developed secure, high-performance FastAPI-based REST APIs with JWT authentication, role-based access control, PostgreSQL integration, and robust backend validations.
- Focused on scalability, data integrity, and clean architecture to support real-world HR operations in multi-organization environments.

Maintenance Assistance (Generative AI)

- Built an advanced RAG pipeline integrated with GPT-4o that unifies equipment manuals, IoT operational data, and maintenance history to deliver AI-powered question answering, task/activity list creation, and remedial actions for alert resolution.
- Empowers maintenance engineers to instantly access accurate, context-aware insights from multiple data sources in one intelligent system.

Multivariate Time Series Anomaly Detection (Predictive Analysis using Deep Learning)

- Developed a zero-labeled anomaly detection system for industrial compressors with an end-to-end pipeline covering time-series preprocessing, model training, and hyperparameter tuning.
- The model leverages Graph Networks, LSTM, and Transformers to capture spatial and temporal relationships, using log-likelihood and Jacobian determinants to learn the latent z-distribution of normal behavior.
- During inference, it compares real-time data against the learned distribution to accurately classify operational states as normal or anomalous.

MOC Project – Inventory Module (Backend Project)

- Designed and implemented the inventory module for Kirloskar's equipment after-sales service, managing warehouse stock availability, daily stock pricing, service ticket generation and bill of materials management.
- Streamlined stock transactions (requisition, outward, inward) and office hierarchy management, leading backend work with minimal senior support.

DM Project – Dubai Municipality (Backend Project)

- Contributed to the Dubai municipality project with diverse use cases, including drainage system monitoring, cooling tower management, recycled water network management and pump monitoring.
- Designed and implemented solutions to optimize these systems, demonstrating strong technical expertise and problem-solving abilities.

KPIX Project (Backend Project)

- Developed a portal for Kpix, the company's flagship project, to display iot device data, including telemetry details, daily/monthly reports, and system updates.
- Implemented user hierarchy and authentication enhancements, enabling multirole support and role-based data access.
- Contributed to features like alert conditioning and widget visualization, demonstrating innovation and technical expertise.

(06/2022 – 01/2023) **Python Django Development Intern – India Khelo Football**

IndiaKheloFootball.com Backend

- Contributed to the development and maintenance of the India Khelo Football website using Django, gaining hands-on experience in backend development and web application architecture.
- Worked extensively with MariaDB for database design, queries, and data management, supporting dynamic website functionalities and performance optimization.

TECHNICAL SKILLS

- **Programming Languages**: Python, C#, HTML, CSS
- **Frameworks** : Fast Api, AI&ML, ASP.NET Core, ASP.net core web api, Flask, Django, Generative AI, AI Agents
- **Database Management**: PostgreSQL, MySQL, Maria DB, SQLite
- **DevOps Tools**: Azure DevOps, Terraform Automation, CI/CD Pipelines, Git, GitHub, Bitbucket
- **IoT Expertise**: IoT data processing, GNSS-based systems, IoT device telemetry, protocol management
- **Tools & Libraries**: Asyncpg, Dapper, ORM, JSON, LINQ, Postman, Swagger, SonarQube, Visual Studio, Visual Studio Code, Google Colab, Jupyter Notebook
- **Azure Certification**: Certified in AZ-900, showcasing expertise in Azure fundamentals and cloud services.
- **IoT Experience**: Contributed to the development of IoT-based products by working on robust backend solutions.

PROFESSIONAL SKILLS

- **System Design**: Developing scalable and efficient backend architectures for IoT solutions
- **API Development**: Designing and implementing restful Api's for diverse applications
- **Team Collaboration**: Cross-functional collaboration with firmware, QA, frontend, and management teams
- **Project Leadership**: Independently driving project modules, including requirement analysis, solution design, and implementation

CERTIFICATIONS

Data Science, AZ-900: Microsoft Azure Fundamentals

EDUCATION

B. Tech (Electrical & Electronics Engineering) - Biju Patnaik University of Technology (2019 - 2023)

LANGUAGES KNOWN

Odia, Hindi, English